Python Programming Foundations

Week 4 – Debugging, Testing, & Reading Structured Code

Day 1: "Debugging as Evidence Gathering"

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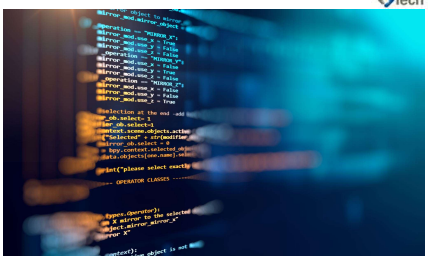

Review Coding Lab

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A Bug Is Information

A bug is information, not a verdict.

When code breaks or gives the wrong answer, the program is giving you clues.

Today, practice debugging by asking:

- What did I expect?
- What actually happened?
- Where is the first useful clue?
- What fix does the evidence justify?

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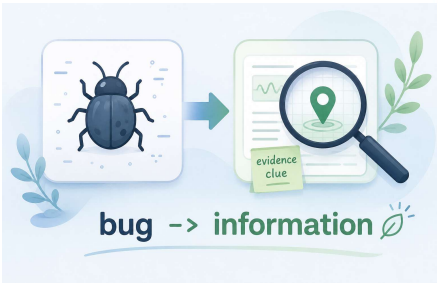

A "bug" is "information"

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Build Becomes Inspect

In earlier weeks, you built small programs.

Now you inspect behavior when a program does not do what it should.

Debugging uses familiar skills:

- reading code
- running code
- tracing values
- explaining output

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Symptoms Show Up, Sources Hide

A *symptom* is what you notice first - The *source* is where the problem actually begins.

Examples:

- **symptom**: wrong final total
- **source**: tax calculated from the wrong value
- **symptom**: program will not run
- **source**: missing quote, colon, or parenthesis

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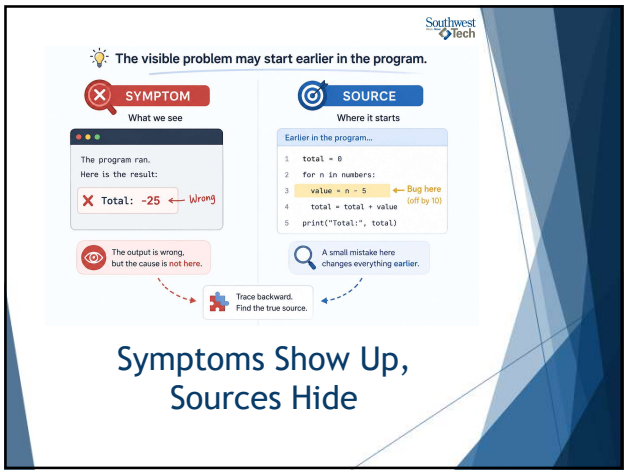
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The visible problem may start earlier in the program.
SYMPTOM
What we see
SOURCE
Where it starts
The program ran.
Here is the result:
Total: -25
Wrong
Earlier in the program...
total = 0
for n in numbers:
value = n - 5
Bug here (off by 10)
total = total + value
print("Total:", total)
The output is wrong, but the cause is not here.
A small mistake here changes everything earlier.
Trace backward. Find the true source.
Symptoms Show Up, Sources Hide

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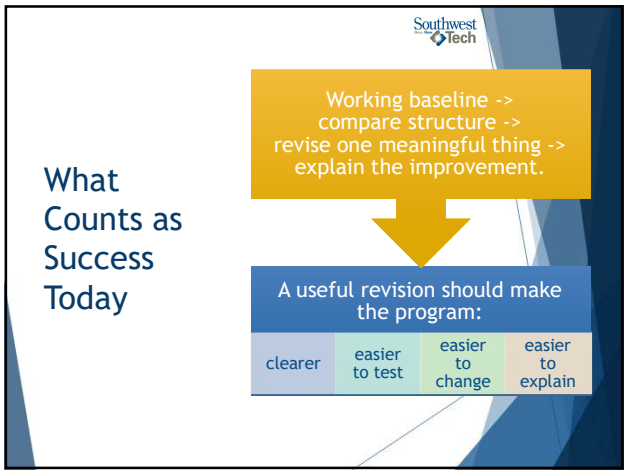
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What Counts as Success Today
Working baseline -> compare structure -> revise one meaningful thing -> explain the improvement.
A useful revision should make the program:
clearer
easier to test
easier to change
easier to explain

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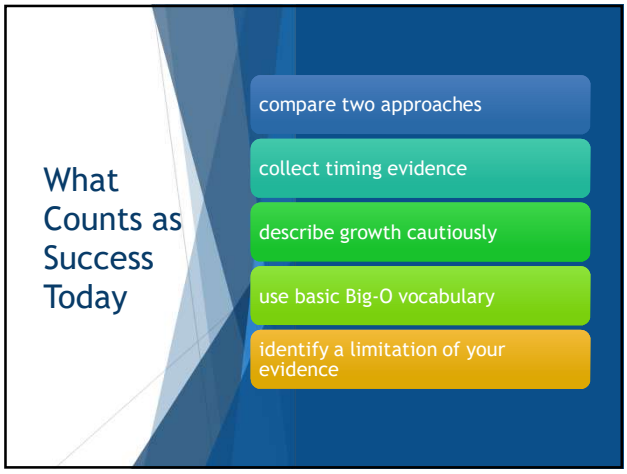
What Counts as Success Today
compare two approaches
collect timing evidence
describe growth cautiously
use basic Big-O vocabulary
identify a limitation of your evidence

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What We Will Use Today
expected versus actual
tracebacks
labeled print-debugging
one-change-at-a-time repair
simple evidence notes
These tools help you investigate before changing code.

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Debugging Toolbox
Small tools. Clear thinking. Better programs.
expected vs actual
traceback clue
labeled print
one-change repair
evidence notes
Debugging is not guessing. It's collecting clues, making small changes, and learning.
Toolbox

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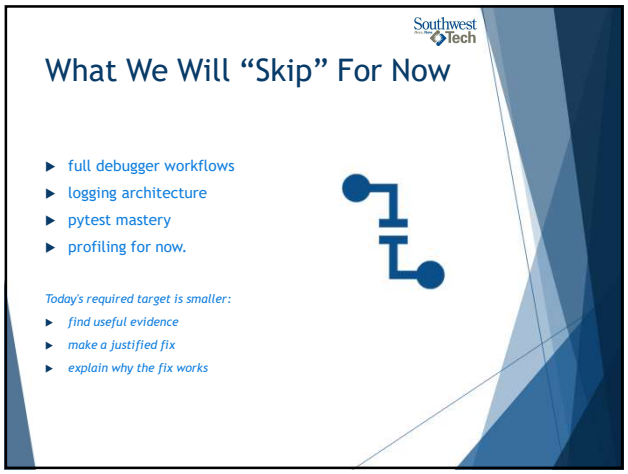
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What We Will "Skip" For Now
full debugger workflows
logging architecture
pytest mastery
profiling for now.
Today's required target is smaller:
find useful evidence
make a justified fix
explain why the fix works

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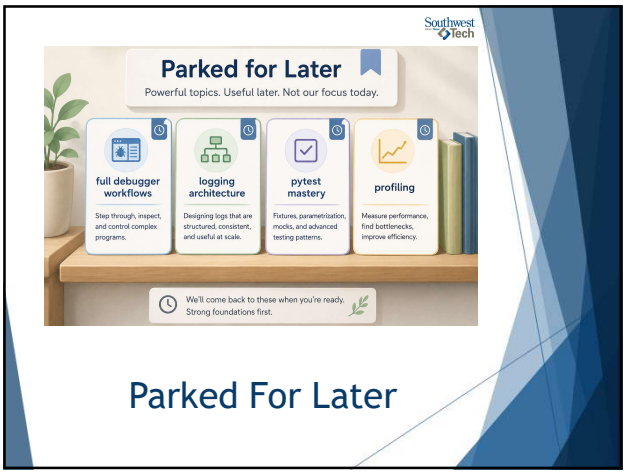

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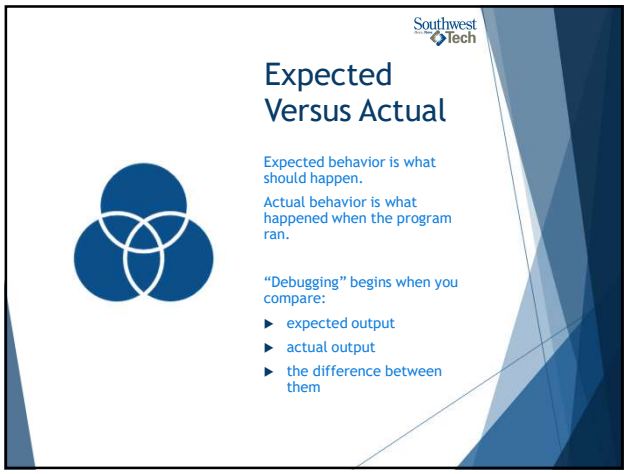

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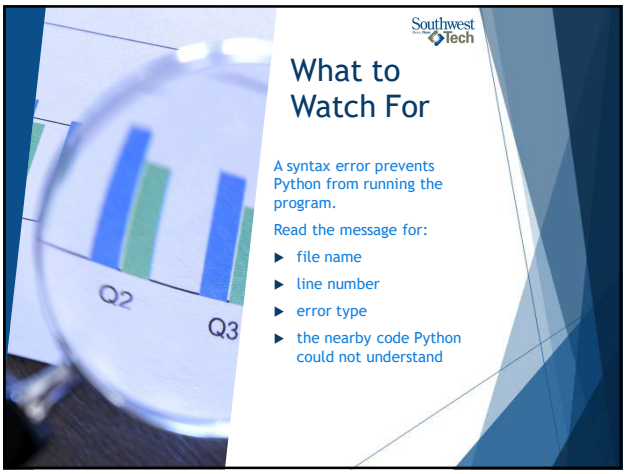

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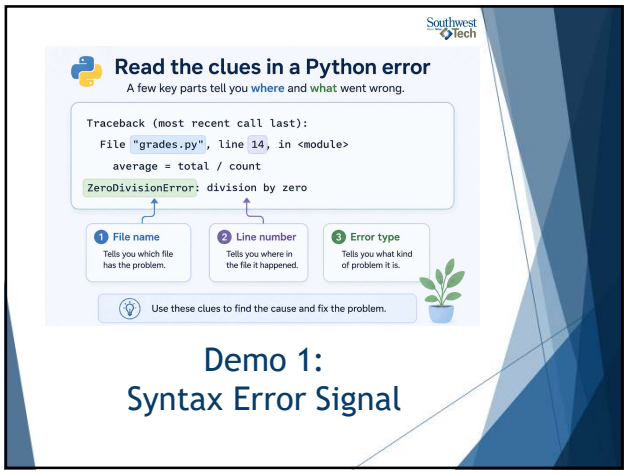

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Demo #2

"Basic Type Awareness"

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What to Watch For

A logic bug can run without crashing and still produce the wrong result.

Trace:

- expected value
- actual value
- suspicious calculation
- corrected calculation
- evidence that the fix worked

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Demo 2: Logic Bug Expected vs Actual

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Ask One Question, Print One Clue

Print-debugging should answer a specific question.

Weak

```
print("here")
```

Better

```
print("subtotal before tax:", subtotal)
```

Every debug print should reveal a value or path you need to inspect.

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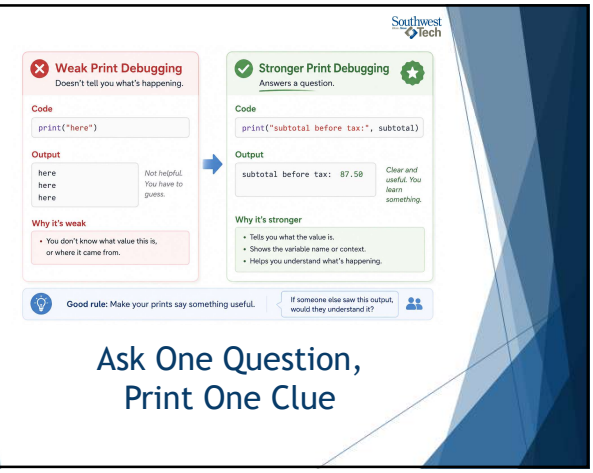

Ask One Question, Print One Clue

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Demo #3

"Grade Summary Debugging"

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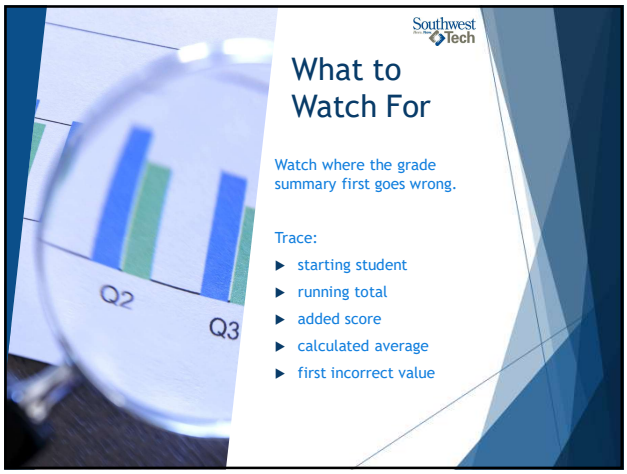

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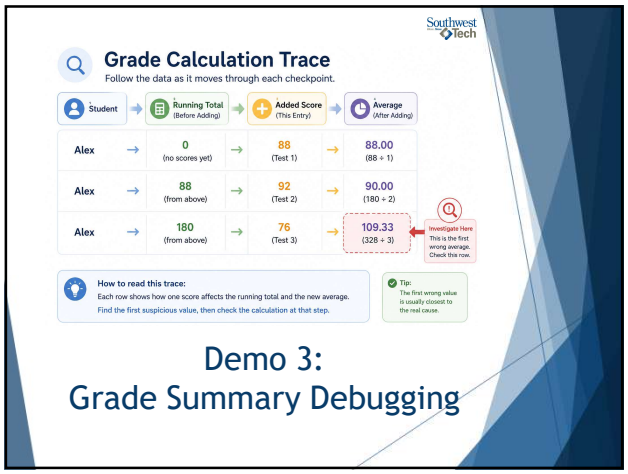

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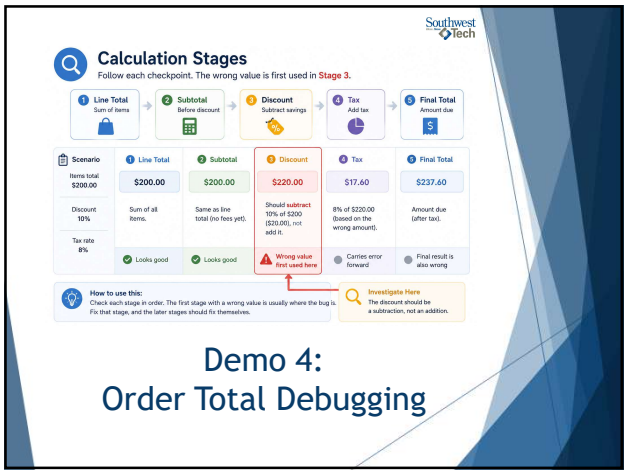

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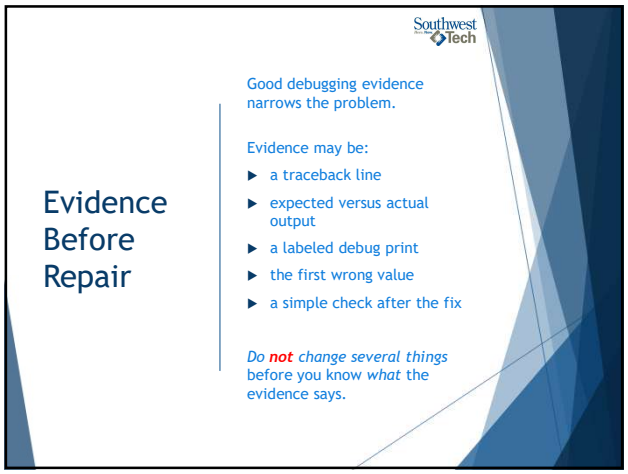

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Common Failure: Changing Code First

Changing code before observing can destroy the trail.

Instead:

- reproduce the problem
- record what happened
- inspect one clue
- change one thing
- run again

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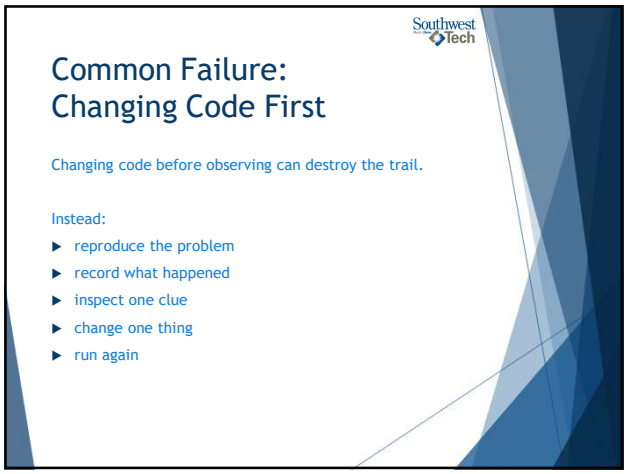

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Assignment 6

For Assignment 6, repair broken code and explain the debugging process.

Your work should show:

- the issue you found
- the evidence that helped
- the fix you made
- the check that proves the fix works

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Evidence For A Debugging Submission

Useful debugging evidence includes:

- corrected `.py` file
- expected-vs-actual comparison
- labeled debug output
- traceback clue
- simple check or test case
- explanation of why the fix works

The evidence should support the repair, not just decorate the submission.

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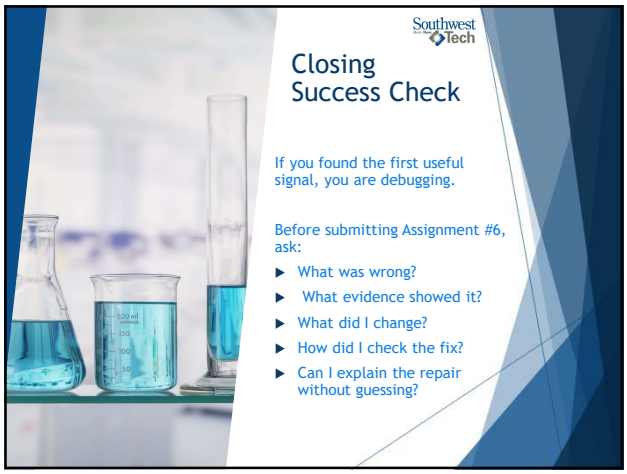
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Closing Success Check
If you found the first useful signal, you are debugging.
Before submitting Assignment #6, ask:
What was wrong?
What evidence showed it?
What did I change?
How did I check the fix?
Can I explain the repair without guessing?

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Thank you!
Have Fun!

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